

Microprocesadores

Prof. Luis Emilio Tonix

HW # 3

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Homework Assignment – ESP32 ADC Single-Shot Moisture Sensor Calibration & Lookup Table vs Regression

Objective

The objective of this experiment was to implement and calibrate a low-cost soil/sand moisture sensor using the ESP32's ADC. By measuring the ADC voltage corresponding to different sand/water proportions, a calibration curve was obtained through polynomial regression. The results were then implemented in two ways:

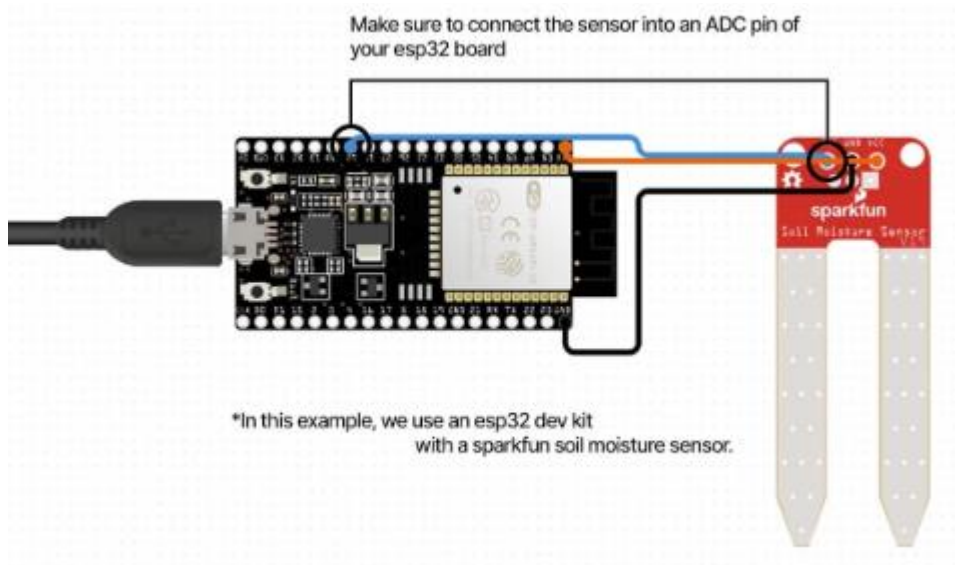
1. **Polynomial Evaluation** – Direct calculation of moisture percentage using a 3rd-degree regression model.
2. **Lookup Table (LUT)** – Precomputed values stored in a header file to map raw ADC readings to moisture percentages in the range 0–100%.

The performance of both methods was compared on the ESP32 in terms of execution time and accuracy.

a) Hardware setup

- Two metallic nails were used as electrodes and inserted into a container filled with sand.
- One nail was connected to **3.3 V** and the other to an **ESP32 ADC pin (GPIO34)** through a **voltage divider**, in order to scale the measured voltage to the proper range of the ESP32's ADC (0–3.3 V).

- A common ground (GND) connection was established between the ESP32 and the container.



b) ESP32 configuration

- The project was based on the official `adc/oneshot_read` example from ESP-IDF.
- The code was adapted to perform periodic measurements (1 sample per second) and display both the raw ADC value and the estimated voltage in the serial monitor.
- ADC calibration was enabled through `adc_cali_raw_to_voltage()` to convert raw readings into millivolts.

Data Collection

- A series of experiments were performed by mixing sand with different proportions of water.
- Voltage readings from the ESP32 ADC were recorded for each water percentage.
- The data was stored in a CSV file, where the **third column corresponded to water percentage** and the **fifth column to measured voltage**.

Data Analysis in Python

- Using Pandas, NumPy, Matplotlib, and Scikit-learn, the data was analyzed.
- A **3rd-degree polynomial regression** was fitted to model the relationship between water percentage and ADC voltage.

- The calibration curve parameters (polynomial coefficients) were extracted for use in C code.
- A **lookup table (LUT)** was generated in Python by evaluating the polynomial across the full ADC range (0–4095) and scaling the results into the range of 0–100%. The LUT was exported as a C header file (lookuptable.h).

ESP32 Implementation

- The ESP32 was programmed to read ADC values using the **ADC oneshot driver**.
- Two functions were implemented:
 - **moisture_map()**: Evaluates the 3rd-degree polynomial for each ADC sample.
 - **lookup_func()**: Uses the LUT to quickly map ADC values to a moisture percentage.
- Both methods were tested on the ESP32, and execution times were measured using the `esp_timer_get_time()` function.
- Continuous measurements were logged, showing side-by-side results for polynomial and LUT methods.

```
Polynomial result = 4.39 % | time = 47 us
Lookup result = 3.00 % | time = 9 us
I (293) MoistureCalibration: ADC Raw: 0 Voltage: 142 mV Moisture: 0.00 %
```

```
I (1293) MoistureCalibration: ADC Raw: 0      Voltage: 142 mV Poly: 0.00 %   LUT: 0.00 %
I (2293) MoistureCalibration: ADC Raw: 0      Voltage: 142 mV Poly: 0.00 %   LUT: 0.00 %
I (3293) MoistureCalibration: ADC Raw: 0      Voltage: 142 mV Poly: 0.00 %   LUT: 0.00 %
I (4293) MoistureCalibration: ADC Raw: 3158   Voltage: 2723 mV      Poly: 76.72 %   LUT: 46.00 %
I (5293) MoistureCalibration: ADC Raw: 3184   Voltage: 2738 mV      Poly: 78.65 %   LUT: 47.00 %
I (6293) MoistureCalibration: ADC Raw: 3191   Voltage: 2742 mV      Poly: 79.18 %   LUT: 47.00 %
I (7293) MoistureCalibration: ADC Raw: 3440   Voltage: 2876 mV      Poly: 99.39 %   LUT: 59.00 %
I (8293) MoistureCalibration: ADC Raw: 4059   Voltage: 3150 mV      Poly: 100.00 %  LUT: 97.00 %
I (9293) MoistureCalibration: ADC Raw: 4031   Voltage: 3139 mV      Poly: 100.00 %  LUT: 95.00 %
I (10293) MoistureCalibration: ADC Raw: 4095  Voltage: 3165 mV      Poly: 100.00 %  LUT: 100.00 %
```

Conclusion

The calibration process successfully modeled the relationship between ADC voltage and moisture percentage using a 3rd-degree polynomial. Results showed that while polynomial regression provides a continuous analytical function, its execution on the ESP32 was slower compared to the LUT approach. In practice:

- The **polynomial method** gave accurate results but required floating-point operations, with execution times around $\sim 47 \mu\text{s}$.
- The **lookup table method** significantly reduced computation time ($\sim 9 \mu\text{s}$) while maintaining comparable accuracy, making it better suited for real-time embedded applications.

Overall, the LUT implementation proved to be the most efficient strategy for deploying the sensor calibration on the ESP32, ensuring fast and consistent readings within the expected 0–100% range. This approach demonstrates that low-cost hardware, combined with proper calibration and optimization, can yield reliable moisture sensing for practical applications in agriculture and environmental monitoring.